

PReLU#

<https://pytorch.org/docs/stable/generated/torch.nn.PReLU.html>

Description:

Applies the element-wise PReLU function. Here `aaa` is a learnable parameter. When called without arguments, `nn.PReLU()` uses a single parameter `aaa` across all input channels. If called with `nn.PReLU(nChannels)`, a separate `aaa` is used for each input channel. weight decay should not be used when learning `aaa` for good performance.

Function Signature

```
class torch.nn.PReLU(num_parameters=1, init=0.25,  
device=None, dtype=None)[source]#
```

Parameters

Shape:

Variables

```
extra_repr()[source]#
```

Returns:

Tensor

Code Examples

```
>>> m = nn.PReLU()  
>>> input = torch.randn(2)  
>>> output = m(input)
```

Notes & Warnings

NOTE Note weight decay should not be used when learning aaa for good performance.

NOTE Note Channel dim is the 2nd dim of input. When input has dims < 2, then there is no channel dim and the number of channels = 1.

Detailed Documentation

Documentation

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Here `aaa` is a learnable parameter. When called without arguments, `nn.PReLU()` uses a single parameter `aaa` across all input channels. If called with `nn.PReLU(nChannels)`, a separate `aaa` is used for each input channel.

weight decay should not be used when learning `aaa` for good performance.

Channel dim is the 2nd dim of input. When input has `dims < 2`, then there is no channel dim and the number of channels = 1.

`num_parameters (int)` – number of `aaa` to learn. Although it takes an `int` as input, there is only two values are legitimate: 1, or the number of channels at input. Default: 1

`init (float)` – the initial value of `aaa`. Default: 0.25

Input: `(*)(*)(*)` where `*` means, any number of additional dimensions.

Output: `(*)(*)(*)`, same shape as the input.

`weight (Tensor)` – the learnable weights of shape `(num_parameters)`.

Return the extra representation of the module.

Runs the forward pass.

Resets parameters based on their initialization used in `__init__`.

